

Srisai Kumar Resu

• srisai.r@protectmymails.com • +1(940)843-1559 •LinkedIn

SUMMARY

Full Stack Software Engineer with 3.5+ years of experience designing and building distributed systems and cloud-native applications for high-volume financial platforms. Strong expertise in backend engineering using Java, Python, and Spring Boot, with deep experience in AWS, Kafka, and microservices architecture. Experienced in data-driven system design and emerging AI technologies, including LLM applications, RAG pipelines, and vector database integration for intelligent enterprise solutions.

SKILLS

Programming Languages: Java, TypeScript, JavaScript, Python
Backend & Distributed Systems: Microservices Architecture, Distributed Systems, RESTful APIs, Apache Kafka, Asynchronous Processing, gRPC
Frontend Development: React.js, Next.js, Angular, HTML5, CSS3, Tailwind CSS
AI / ML & GenAI: LLM, Generative AI, RAG (Retrieval-Augmented Generation), OpenAI APIs, Hugging Face, Vector Databases (Pinecone, FAISS)
Cloud & DevOps: AWS (EKS, EC2, S3, Lambda, API Gateway), Docker, Kubernetes, Terraform, CI/CD Pipelines, Jenkins, GitHub Actions
Databases & Data Engineering: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, SQL Optimization, Indexing
System Design & Architecture: System Design, Cloud-Native Architecture, Scalable Systems, Fault-Tolerant Systems, High Availability Systems
Observability & Monitoring: Prometheus, Grafana, OpenTelemetry, Datadog, CloudWatch
Security & API Management: OAuth2, JWT, RBAC, API Gateway, Secure API Design, Input Validation
Testing & Quality Engineering: JUnit, Mockito, Jest, Cypress, Selenium, Integration Testing
Collaboration & Agile: Agile (Scrum/Kanban), JIRA, Confluence, Code Reviews, Sprint Planning

PROFESSIONAL EXPERIENCE

Paypal
Software Engineer

Jun 2025 – Present | Austin, Texas

- Built a smart payment routing solution using Spring Boot microservices that dynamically selected payment paths based on cost efficiency and authorization success metrics, resulting in a 9% increase in payment approvals and sub-60ms transaction latency.
- Designed resilient distributed workflows using fault-tolerant patterns including circuit breakers, retries, and bulkhead isolation with Resilience4j, improving service stability and reducing payment failures by 40% during high-traffic processing periods.
- Implemented high-throughput REST APIs with idempotency guarantees using Redis-based request deduplication and distributed locking mechanisms, enabling consistent transaction processing across services handling 15K+ concurrent payment requests.
- Engineered real-time transaction monitoring dashboards using React and Spring MVC with optimized API requests, caching strategies, and asynchronous polling, reducing dashboard load times by 35% and improving operational system visibility.
- Optimized PostgreSQL database performance through advanced indexing strategies, query plan analysis, and table partitioning techniques, increasing transaction throughput by 30% and improving query execution efficiency by 45%.
- Deployed cloud-native microservices using AWS ECS Fargate, Lambda, API Gateway, and RDS, enabling elastic scaling, fault-tolerant processing, and 99.95% platform availability across mission-critical financial transaction systems.
- Implemented secure API access using OAuth2/JWT authentication and RBAC authorization through Spring Security, protecting internal and external payment APIs across distributed microservices within enterprise production environments.
- Orchestrated an internal AI-powered knowledge assistant using OpenAI APIs, LangChain, and Pinecone vector search, enabling rapid access to API specifications, operational runbooks, and service documentation across payment platform engineering teams.
- Automated CI/CD pipelines using Jenkins and GitHub Actions integrated with Datadog and CloudWatch monitoring, reducing deployment time by 38% and improving mean time to recovery for production incidents.

Capgemini
Software Engineer

May 2021– Dec 2023 | Mumbai, India

- Developed high-performance backend microservices using Java and Spring Boot for partner onboarding, payment integration, and transaction workflows, supporting 3M+ annual transactions across distributed financial systems with optimized API performance and reliable execution.
- Constructed web applications using Angular and TypeScript, developing reusable components, optimized state management, and responsive user interfaces that reduced page load times by 28% and improved application performance across business-critical workflows.
- Automated cloud environment provisioning using Docker and Infrastructure-as-Code practices, reducing environment setup time by 70% and improving deployment consistency across development and testing environments.
- Optimized MySQL database performance through indexing strategies, query optimization, and Redis caching layers, improving API response times by 35% and reducing database query latency across high-volume reporting and financial processing modules.
- Implemented API Gateway policies, request validation, audit logging, and secrets management controls across financial services, strengthening compliance requirements and improving visibility into API usage patterns.
- Operationalized centralized observability solutions using application logs, performance metrics, and real-time monitoring dashboards, reducing troubleshooting effort and accelerating issue diagnosis across distributed services.
- Constructed Spring Boot microservices architecture enabling independent service deployments and improved service isolation, reducing system downtime by 25% while supporting modular development across multiple financial processing services.
- Architected event-driven backend services using Apache Kafka for asynchronous communication and reliable data synchronization between distributed systems, improving processing throughput by 2.5× across transaction processing pipelines.
- Designed end-to-end system architecture for distributed financial transaction services using system design principles including microservices decomposition, API contracts, event-driven workflows, scalability, and fault tolerance to support high-throughput payment processing systems.
- Architected automated testing frameworks using JUnit, Mockito, Jest, and Cypress, increasing overall test coverage to 85%+ and reducing production defects by 30% through improved integration and end-to-end testing strategies.

ACADEMIC PROJECTS

AI-Powered Smart Code Reviewer & Debug Assistant

- Configured a GenAI-assisted code review tool using React, FastAPI, and GPT APIs that analyzes user-submitted code to identify potential bugs, explain issues in natural language, and suggest context-aware fixes, improving code debugging efficiency for learners and developers.
- Programmed an LLM-powered code analysis pipeline with structured prompting to generate unit test suggestions and basic code optimizations (time/space improvements), enabling faster validation of edge cases and better code quality during development workflows.

AI-Powered E-Commerce Product Recommendation & Search Platform

- Engineered a full-stack e-commerce platform using React and Spring Boot, enabling product catalog management, user tracking, and end-to-end purchase workflows with optimized backend APIs and seamless user experience.
- Built an AI-driven recommendation engine using OpenAI APIs and lightweight machine learning models to generate personalized product suggestions based on user browsing and purchase history.

EDUCATION

Master of Science in Data Science | University of North Texas — Denton, Texas